



Mata Kuliah : Pemrograman Framework
Program Studi : D4 – Teknik Informatika
Semester : 6

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Jobsheet Ke- : 18

Laporan Jobsheet

Praktikum 1

1.

```

C:\Downloads\git\PemrogramanFramework\pertemuan02\my-app → (P main) 22.22.1 1ms 5:09 PM
• s fandy >> npm install jest-jest-environment-jsdom @testing-library/react @testing-library/jest-dom --save-dev
--force
npm WARN deprecated inflight@1.0.6: This module is not supported, and leaks memory. Do not use it. Check out lru
-cache if you want a good and tested way to coalesce async requests by a key value, which is much more comprehen
sive and powerful.
npm WARN deprecated whatwg-encoding@3.1.1: Use @exodus/bytes instead for a more spec-conformant and faster imple
mentation
npm WARN deprecated glob@10.5.0: Old versions of glob are not supported, and contain widely publicized security
vulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be pu
rchased (at exorbitant rates) by contacting i@izs.me
npm WARN deprecated glob@7.2.3: Old versions of glob are not supported, and contain widely publicized security v
ulnerabilities, which have been fixed in the current version. Please update. Support for old versions may be pur
chased (at exorbitant rates) by contacting i@izs.me

added 286 packages, and audited 782 packages in 14s

198 packages are looking for funding
  run `npm fund` for details

2 vulnerabilities (1 high, 1 critical)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
C:\Downloads\git\PemrogramanFramework\pertemuan02\my-app → (P main) 22.22.1 14.456s 5:09 PM
• s fandy >> npm audit fix --force
npm WARN using --force Recommended protections disabled.

changed 5 packages, and audited 782 packages in 15s

198 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
C:\Downloads\git\PemrogramanFramework\pertemuan02\my-app → (P main) 22.22.1 14.794s 5:10 PM
• s fandy >>
```

```

JS jest.config.mjs > ...
1 import nextJest from 'next/jest.js'
2
3 const createJestConfig = nextJest({
4   dir: './',
5 })
6
7 const config = {
8   coverageProvider: 'v8',
9   testEnvironment: 'jsdom',
10 }
11
12
13 export default createJestConfig(config)
```



```
{ } package.json > { } scripts > abc test:watch
1  {
2    "name": "my-app",
3    "version": "0.1.0",
4    "private": true,
5    "scripts": {
6      "dev": "next dev",
7      "build": "next build",
8      "start": "next start",
9      "lint": "next lint",
10     "test": "jest --passWithNoTests -u",
11     "test:coverage": "npm run test -- --coverage",
12     "test:watch": "npm run test -- --watch"
13   },
14   "dependencies": {
15     "@types/node": "25.2.2"
```

- Menambahkan element testing menggunakan jest testing

```
✓ MY-APP
> .next
> node_modules
> public
✓ src
  ✓ _test_/components
    > components
    > Middleware
    > pages
    > styles
    > types
```

2.

- Membuat file *test/components*



```
src > _test_ > pages > about.spec.tsx > describe("About Page") callback > it("renders  
1  import {render} from "@testing-library/react"  
2  import About from "@pages/about"  
3  
4  describe("About Page", () => {  
5    it("renders the about page correctly", () => {  
6      const page = render(<About />)  
7      expect(page).toMatchSnapshot()  
8    })  
9  })
```

3.

```
... \Downloads\git\PemrogramanFramework\pertemuan02\my-app → ( ?  
$fandy >> npm run test  
PASS src/_test_/pages/about.spec.tsx  
  About Page  
    ✓ renders the about page correctly (13 ms)  
  
  > 1 snapshot written.  
Snapshot Summary  
  > 1 snapshot written from 1 test suite.  
  
Test Suites: 1 passed, 1 total  
Tests:       1 passed, 1 total  
Snapshots:  1 written, 1 total  
Time:        0.534 s  
Ran all test suites.  
... \Downloads\git\PemrogramanFramework\pertemuan02\my-app → ( ? mai  
o $fandy >> |
```

- Melakukan test menggunakan halaman atribut about



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
...Downloads\git\PemrogramanFramework\pertemuan02\my-app → (main) 22.22.1 1.459s
fandy >> npm run test:coverage

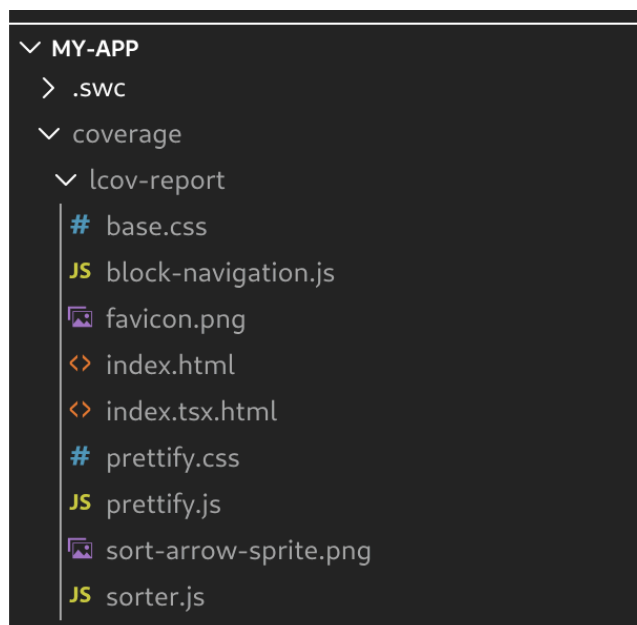
> my-app@0.1.0 test:coverage
> npm run test -- --coverage

> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

PASS src/_test_/pages/about.spec.tsx
  About Page
    ✓ renders the about page correctly (14 ms)

-----|-----|-----|-----|-----|
File    | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-----|-----|-----|-----|-----|
All files|    100 |    100   |    100   |    100   |
index.tsx|    100 |    100   |    100   |    100   |
-----|-----|-----|-----|-----|
Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
Snapshots:   1 passed, 1 total
Time:        0.661 s, estimated 1 s
Ran all test suites.
...Downloads\git\PemrogramanFramework\pertemuan02\my-app → (main) 22.22.1 1.668s
fandy >>
```

4.



- Melakukan test coverage



5.

```
Welcome to fish, the friendly interactive shell
Type help for instructions on how to use fish

fandy >> cd Downloads/giti/PemrogramanFramework/pertemuan02/my-app/
fandy >> npm run test:coverage

my-app@0.1.0 test:coverage
npm run test -- --coverage

my-app@0.1.0 test
jest --passWithNoTests -u --coverage

PASS src/_test_/pages/about.spec.tsx
  About Page
    ✓ renders the about page correctly (13 ms)

File                                     % Stmts   % Branch  % Funcs   % Lines  Uncovered Line #s
-----
All files                                1.6         0      2.27     1.73
my-app                                  100       100      100       100
next-env.d.ts                          0         0        0         0       3
my-app/src                             100       100      100       100
  middleware.ts                        100       100      100       100
  my-app/src/Middleware                 100       100      100       100
  withAuth.ts                          100       100      100       100
  .../src/components/Layouts/AppShell  100       100      100       100
  index.tsx                            100       100      100       100
  ...pp/src/components/Layouts/navbar  100       100      100       100
  index.tsx                            100       100      100       100
my-app/src/pages                       100       100      100       100
  404.tsx                              100       100      100       100
  app.tsx                              100       100      100       100
  document.tsx                         100       100      100       100
  index.tsx                            100       100      100       100
my-app/src/pages/about                 100       100      100       100
```

Code coverage report for All files...

File	Statements	Branches	Functions
my-app	0%	0/1	100%
my-app/src	0%	0/8	100%
my-app/src/Middleware	0%	0/22	0%
my-app/src/components/Layouts/AppShell	0%	0/9	0%
my-app/src/components/Layouts/navbar	0%	0/8	0%
my-app/src/pages	0%	0/12	100%
my-app/src/pages/about	100%	3/3	100%
my-app/src/pages/admin	0%	0/1	100%
my-app/src/pages/auth	0%	0/2	100%
my-app/src/pages/blog	0%	0/5	100%
my-app/src/pages/editor	0%	0/1	100%
my-app/src/pages/produk	0%	0/41	0%
my-app/src/pages/profile	0%	0/3	100%
my-app/src/pages/profile/edit	0%	0/1	100%
my-app/src/pages/setting	0%	0/3	100%
my-app/src/pages/setting/user/password	0%	0/3	100%
my-app/src/pages/user	0%	0/3	100%
my-app/src/utils/db	0%	0/56	0%

- Melakukan tes secara keseluruhan dengan menggunakan coverage dengan hasil



6.

```
src > pages > about > index.tsx > ...
1  import Head from 'next/head'
2  import Image from 'next/image'
3  import { Inter } from 'next/font/google'
4  import styles from '@styles/Home.module.css'
5
6  const inter = Inter({ subsets: ['latin'] })
7
8  export default function Home() {
9    return (
10     <div>
11       <h1 data-testid="title">ini halaman about</h1>
12     </div>
13   )
14 }
15
```

```
src > _test_ > pages > about.spec.tsx > ...
1  import {render} from "@testing-library/react"
2  import About from "@pages/about"
3
4  describe("About Page", () => {
5    it("renders the about page correctly", () => {
6      const page = render(<About />)
7      expect(page.getByTestId("title").textContent).toBe("ini halaman about")
8      expect(page).toMatchSnapshot()
9    })
10 })
```

document.tsx	0	100	0	0	1-3
index.tsx	0	100	0	0	2
my-app/src/pages/about	100	100	100	100	
index.tsx	100	100	100	100	
my-app/src/pages/admin	0	100	0	0	

```
about.spec.tsx U JS jest.config.mjs U index.tsx M
src > _test_ > pages > about.spec.tsx > describe("About Page") callback > it("renders the about page correctly") callback
1  import {render} from "@testing-library/react"
2  import About from "@pages/about"
3
4  describe("About Page", () => {
5    it("renders the about page correctly", () => {
6      const page = render(<About />)
7      expect(page.getByTestId("title").textContent).toBe("about")
8      expect(page).toMatchSnapshot()
9    })
10 })
```



```
🔍 dock ➔ Downloads\git\PemrogramanFramework\pertemuan02\my-app ➔ (🔄 main) @ 22.2
⚡ fancy >> npm run test:coverage
> my-app@0.1.0 test:coverage
> npm run test -- --coverage

> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage

FAIL src/_test_/pages/about.spec.tsx 10/10
  About Page
    ✕ renders the about page correctly (15 ms)

● About Page › renders the about page correctly

expect(received).toBe(expected) // Object.is equality

Expected: "about"
Received: "ini halaman about"

   5 |         it("renders the about page correctly", () => {
>         const page = render(<About />)
      |                           ^
      |                           expect(page.getByTestId("title").textContent).toBe("about")
      |                           expect(page).toMatchSnapshot()
      |                           })
      |                           })
      |                           at Object.toBe (src/_test_/pages/about.spec.tsx:7:55)
```

- Mengubah atribut dalam tes sehingga menghasilkan halaman yang diperlukan dan melakukan testing jika case fail

```
about.spec.tsx U product.spec.tsx U JS jest.config.mjs U
src > _test_ > pages > product.spec.tsx > describe("About Page") callback > it("renders the about page correctly")

1  import {render, screen} from "@testing-library/react"
2  import Produk from "@pages/produk"
3
4  describe("About Page", () => {
5    it("renders the about page correctly", () => {
6      const page = render(<Produk />)
7      expect(page.getByTestId("title").textContent).toBe("Product Page")
8      expect(page).toMatchSnapshot()
9    })
10 })
```

7.



```
New Tab Split View
dock → Downloads\git\PemrogramanFramework\pertemuan02\my-app → (main) @ 22.22.1 1ms 6:
fandy >> npm run test:coverage
my-app@0.1.0 test:coverage
npm run test -- --coverage

my-app@0.1.0 test
jest --passWithNoTests -u --coverage

PASS src/_test_/pages/about.spec.tsx
FAIL src/_test_/pages/product.spec.tsx
  ● About Page › renders the about page correctly
    NextRouter was not mounted. https://nextjs.org/docs/messages/next-router-not-mounted

  7 | export default function kategori() {
    |                                     ^
  8 |   const { login, logout } = useState(false);
    |                                     ^
  9 |   const { push } = useRouter();
    |                                     ^
 10 |   const [products, setProducts] = useState([]);
    |                                     ^
 11 |
    |                                     ^
 12 |   // Gunakan destructuring untuk mengambil data dari state
```

```
about.spec.tsx U product.spec.tsx U X JS jest.config.mjs U
src > _test_ > pages > product.spec.tsx > jest.mock("next/router") callback > useRouter

1 import { render, screen } from "@testing-library/react"
2 import Produk from "@pages/produk"
3 import { use } from "react"
4 import { on } from "node:cluster"
5
6 jest.mock("next/router", () => ({
7   useRouter() {
8     return {
9       route: "/produk",
10      pathname: "",
11      query: {},
12      asPath: "",
13      push: jest.fn(),
14      event: {
15        on: jest.fn(),
16        off: jest.fn(),
17      },
18      isReady: true,
19    }
20  }
21 })))
22
```

- Melakukan testing pada halaman produk dan menambahkan jest mock agar dapat melakukan mounting



8.

```
❗ fandy >> npm run test:coverage
> my-app@0.1.0 test:coverage
> npm run test -- --coverage

PRAKTIKUM 8 – Menangani Undefined Data
> my-app@0.1.0 test
  * Jalankan npm run test:coverage maka akan muncul error
> jest --passWithNoTests -u --coverage

FAIL src/_test_/pages/product.spec.tsx
  ● About Page › renders the about page correctly

  TypeError: Cannot read properties of undefined (reading 'data')

    39 |         <main>
    40 |           { /* Logika pengiriman data: jika loading kirim array kosong untuk skeleton */ }
    41 |           <TampilanProduk products={data.data} />
    42 |         </main>
    43 |       </div>
    44 |     );
    45 |   });
    46 | }
    47 |
    48 | at data (src/pages/produk/index.tsx:41:46)
    49 | at Object.react_stack_bottom_frame (node_modules/react-dom/cjs/react-dom-client.development.js:25904:20)
    50 | at renderWithHooks (node_modules/react-dom/cjs/react-dom-client.development.js:7662:22)
    51 | at updateFunctionComponent (node_modules/react-dom/cjs/react-dom-client.development.js:10166:19)
    52 | at beginWork (node_modules/react-dom/cjs/react-dom-client.development.js:11778:18)
    53 | at runWithFiberInDEV (node_modules/react-dom/cjs/react-dom-client.development.js:874:13)
    54 | at performUnitOfWork (node_modules/react-dom/cjs/react-dom-client.development.js:17641:22)
    55 | at workLoopSync (node_modules/react-dom/cjs/react-dom-client.development.js:17469:41)
    56 | at renderRootSync (node_modules/react-dom/cjs/react-dom-client.development.js:17450:11)
    57 | at performWorkOnRoot (node_modules/react-dom/cjs/react-dom-client.development.js:16583:35)
    58 | at performWorkOnRootViaSchedulerTask (node_modules/react-dom/cjs/react-dom-client.development.js:18957:7)
    59 | at flushActQueue (node_modules/react/cjs/react.development.js:590:34)
    60 | at process.env.NODE_ENV.exports.act (node_modules/react/cjs/react.development.js:884:10)
    61 | at node_modules/@testing-library/react/dist/act-compat.js:46:25
    62 | at renderRoot (node_modules/@testing-library/react/dist/pure.js:189:26)
```

```
39 |         <main>
40 |           { /* Logika pengiriman data: jika loading kirim array kosong untuk skeleton */ }
41 |           <TampilanProduk products={isLoading ? [] : data?.data} />
42 |         </main>
43 |       </div>
```

```
src > _test_ > pages > ⚙ about.spec.tsx > ...
1  import {render} from "@testing-library/react"
2  import About from "@pages/about"
3
4  describe("About Page", () => {
5    it("renders the about page correctly", () => {
6      const page = render(<About />)
7      // expect(page.getByTestId("title").textContent).toBe("ini halaman about")
8      expect(page).toMatchSnapshot()
9    })
10 })
```

```
23 describe("About Page", () => {
24   it("renders the about page correctly", () => {
25     const page = render(<Produk />)
26     // expect(page.getByTestId("title").textContent).toBe("Daftar Produk")
27     expect(page).toMatchSnapshot()
28   })
29 })
```



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Apr 22 18:47

Jobsheet - Google Dr... 11_Fandy Wahyu Han... DJ HOTEL ROOM... Next.js Routing for C... Code coverage repor...

127.0.0.1:3000/coverage/lcov-report/index.html?vscode-livepreview=true

All files

8.55% Statements 16/187 14.54% Branches 8/55 6.81% Functions 3/44 8.09% Lines 14/173

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File	Statements	Branches	Functions	Lines
my-app	0%	0/1	100%	0/0
my-app/src	0%	0/8	100%	0/0
my-app/src/Middleware	0%	0/22	0%	0/2
my-app/src/components/layouts/AppShell	0%	0/9	0%	0/3
my-app/src/components/layouts/navbar	0%	0/8	0%	0/3
my-app/src/pages	0%	0/12	100%	0/0
my-app/src/pages/about	100%	3/3	100%	1/1
my-app/src/pages/admin	0%	0/1	100%	0/0
my-app/src/pages/auth	0%	0/2	100%	0/0
my-app/src/pages/blog	0%	0/5	100%	0/0
my-app/src/pages/editor	0%	0/1	100%	0/0
my-app/src/pages/produk	21.95%	9/41	80%	8/10
my-app/src/pages/profile	0%	0/3	100%	0/0

- melakukan unit testing pada bagian produk dan memiliki angka 8.55 persen coverage



Tugas Praktikum

1. tugas 1:

```
my-app: fish — Konsole
New Tab Split View
dock fandy >> npm run test:coverage
> my-app@0.1.0 test:coverage
> npm run test -- --coverage
> my-app@0.1.0 test
> jest --passWithNoTests -u --coverage
PASS src/_test_/pages/product.spec.tsx
PASS src/_test_/pages/about.spec.tsx
-----
File      % Stmts   % Branch   % Funcs   % Lines   Uncovered Line #s
-----
All files    90.47     90.9      71.42     90
pages/about    100      100      100      100
index.tsx      100      100      100      100
pages/produk    90       100       50       90
index.tsx      90       100       50       90
views/produk   87.5     66.66     75      85.71
index.tsx      87.5     66.66     75      85.71
-----
Test Suites: 2 passed, 2 total
Tests:       3 passed, 3 total
Snapshots:  1 passed, 1 total
Time:        0.801 s, estimated 1 s
Ran all test suites.
```

127.0.0.1:3000/coverage/lcov-report/index.html?vscode-livepreview=true

All files
90.47% Statements 19/21 90.9% Branches 18/11 71.42% Functions 5/7 90% Lines 18/20

Press n or j to go to the next uncovered block, b, p or k for the previous block.

Filter:

File	Statements	Branches	Functions	Lines
pages/about	100%	3/3	100%	3/3
pages/produk	90%	9/10	100%	9/10
views/produk	87.5%	7/8	66.66%	6/7

Code coverage generated by Istanbul at 2026-04-22T12:39:55.559Z



Pertanyaan Refleksi

1. Mengapa unit testing penting sebelum production?

Unit testing merupakan fondasi utama dalam menjaga kualitas perangkat lunak sebelum dirilis ke pengguna akhir.

- **Deteksi Dini Bug:** Memungkinkan pengembang menemukan kesalahan pada level fungsi terkecil, sehingga mencegah akumulasi *technical debt* yang lebih kompleks.
- **Keamanan dalam Refactoring:** Memberikan jaminan bahwa perubahan kode atau optimasi tidak merusak fungsionalitas yang sudah ada (*regression testing*).
- **Efisiensi Biaya:** Mengidentifikasi dan memperbaiki kesalahan pada tahap pengembangan jauh lebih ekonomis dibandingkan menangani kegagalan sistem saat aplikasi sudah di tangan pengguna.

2. Mengapa branch coverage sulit mencapai 100%?

Branch coverage mengukur sejauh mana semua jalur percabangan logika (seperti **if-else** atau **switch-case**) telah dieksekusi oleh tes.

- **Unreachable Code:** Terdapat blok kode tertentu, seperti penanganan kesalahan sistem yang sangat spesifik (*exception handling*), yang sangat sulit atau bahkan mustahil untuk disimulasikan dalam lingkungan pengujian.
- **Kompleksitas Logika:** Semakin banyak percabangan bersarang (*nested logic*), semakin banyak skenario unik yang harus dibuat, yang terkadang tidak sebanding dengan nilai tambah yang dihasilkan.
- **Dependensi Eksternal:** Logika yang bergantung pada pustaka pihak ketiga seringkali hanya dapat diuji pada jalur sukses saja, sementara jalur kegagalan internal pustaka tersebut berada di luar kendali pengujian unit.

3. Apa itu mocking?



Mocking adalah teknik dalam unit testing untuk mengganti dependensi asli (seperti API, database, atau modul router) dengan objek tiruan yang mensimulasikan perilaku modul tersebut.

- **Tujuan Utama:** Mengisolasi unit yang sedang diuji agar tidak terpengaruh oleh faktor luar yang tidak stabil, lambat, atau memiliki akses terbatas.
- **Implementasi:** Dalam praktikum ini, penggunaan `jest.mock` pada SWR dan Next Router memungkinkan pengujian komponen tetap berjalan secara konsisten tanpa memerlukan koneksi internet atau server backend yang aktif.

4. Kapan snapshot test digunakan?

Snapshot test digunakan ketika pengembang ingin memastikan bahwa struktur **User Interface (UI)** tidak berubah secara tidak sengaja.

- **Mekanisme:** Jest mengambil representasi struktur DOM komponen dan menyimpannya sebagai referensi. Pada pengujian berikutnya, Jest akan membandingkan output saat ini dengan referensi tersebut.
- **Kesesuaian:** Sangat efektif digunakan pada komponen yang memiliki tampilan visual stabil dan jarang mengalami perubahan struktur elemen yang drastis.

5. Apakah semua file harus dites?

Secara pragmatis, **tidak semua file harus diuji**.

- **Prioritas:** Pengujian harus difokuskan pada file yang mengandung logika bisnis kritis (*business logic*), pemrosesan data, dan fitur-fitur utama yang berdampak langsung pada pengguna.
- **Pengecualian:** File konfigurasi (seperti `next.config.js`), file definisi tipe data (`.d.ts`), serta file *boilerplate* atau *styling* statis tidak memerlukan unit test karena tidak memiliki alur logika fungsional yang dinamis.



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- **Optimalisasi:** Strategi terbaik adalah mencapai cakupan yang tinggi pada modul-modul esensial daripada memaksakan cakupan 100% pada seluruh proyek yang justru dapat menghambat efisiensi pengembangan.